Rudransh Singh

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EDUCATION

University of California, Berkeley May 2028

Bachelor of Science, Electrical Engineering & Computer Science

Relevant Coursework: Data Structures and Algorithms (CS 61B), Discrete Math and Probability Theory (CS 70), Linear Algebra (Math 110), Optimization Models in Engineering (EECS 127), Intro to Signals (EECS 16A)

EXPERIENCE

Co-Founder | InsightLabAI March 2026 - August 2026

- Built a synthetic user research platform that simulates how customers respond to new product features, using large language model driven digital twins in Python.
- Designed a calibration pipeline that captures prediction and outcome pairs from live customer pilots and scores simulated persona responses against observed user behavior.
- Scaled to 10k MRR and 20 paying companies.
- Backed by SkyDeck and Z-Fellows; Raised 210k

Software Engineering Intern | Nuntius (YC S25) May 2025 - August 2025

- Developed an evaluation harness that scored model responses against policy specifications, reporting rule-violation rates across prompt and model variants to guide release decisions.
- Implemented guardrail checks in the inference pipeline that validated and repaired responses against constraint schemas before they were returned to clients.
- Added regression testing and logging over evaluation suites so changes in rule-compliance were caught before deployment.

PROJECTS

InsightLabAI

- Architected an asynchronous inference orchestration layer over FastAPI and asyncio that fans each research query out to 20 persona simulations, using asyncio.Semaphore(8) to cap in-flight provider calls and prevent rate-limit cascades under burst load.
- Hardened the request path with exponential backoff and jitter, per-call timeouts, and circuit breaking after repeated provider failures, holding p99 latency at 21 seconds across 10 concurrent research sessions.

RESEARCH AND PUBLICATIONS

Machine Learning Research Summer 2022 - February 2025

- Rana, A., Singh,R., et al.. “Enhancing Electronic Health Record Systems with Serverless Blockchain Integration”. CCCCI 2024.
- Built Solidity smart contracts for on-chain EHR access control and co-designed a serverless Ethereum and Azure Functions architecture that replaces always-on node infrastructure with on-demand compute.
- Singh, R., et al. “Identifying Race and Gender Bias in Stable Diffusion AI Image Generation.” ICAIC 2024.
- Audited 358 Stable Diffusion generations for demographic skew using CLIP-based classification, quantifying representation gaps across occupation prompts.
- Singh, R., et al. “Stance Detection of Political Tweets with Transformer Architectures.” ICTC, 2022.
- Fine-tuned BERT and RoBERTa transformer classifiers on 4300 labeled tweets for stance detection, reaching 87% F1; built the annotation and evaluation pipeline in PyTorch.

SKILLS

Languages: Python, Java, SQL, JavaScript
Frameworks: React, Node.js, FastAPI, LangChain, Git, PostgreSQL
ML Libraries: TensorFlow, PyTorch, Pandas, NumPy